Bridging Cryptographic Trust, Operational Risk, and Long-Term Security in Low-Latency Critical Systems

Sven Wohlgemuth[1,a)] Kazuo Takaragi[2,b)] Katsuyuki Umezawa[3,c)] Kazuo Furukawa[4,d)]

Abstract: Low-Latency Critical Systems, including Cyber-Physical Systems, high-frequency trading, energy control, and distributed robotics, require secure trust management under operational uncertainty and long-term cyber risk. This paper proposes SK4SC (Secure Kernel for Supply Chains), a digital trust framework that models digital identity usage as a micro-transaction, enabling explicit evaluation of benefit, cost, and risk. The framework interprets revocation as a mechanism for balancing Type I (false rejection) and Type II (false acceptance) errors through adaptive hybrid revocation. It unifies the cryptographic trust model with context-, time-, and behavior-aware trust models. SK4SC provides cryptographic evidence on transactions by integrating privacy-preserving authentication, ledger-based auditability, post-quantum cryptography, and IK-based communication mechanisms toward information-theoretic confidentiality. While post-quantum mechanisms provide protection against known quantum attacks, the IK approach reduces dependence on long-term computational assumptions and mitigates harvest-now-decrypt-later risks. The proposed framework provides a consistent foundation for informatics as a scientific tool for the open information economy.

Keywords: Digital Trust, Hybrid Revocation, Low-Latency Critical Systems, Post-Quantum Cryptography, Long-Term Security

1. Introduction

Beyond 5G/6G networks are expected to support a wide range of Low-Latency Critical Systems (LLCS), including Cyber-Physical Systems (CPS), high-frequency trading, energy control, distributed robotics, and digital supply chains. These environments require continuous interaction among heterogeneous entities under stringent latency constraints, making secure and privacy-preserving digital trust management a fundamental requirement.

Conventional trust infrastructures based on Public Key Infrastructure (PKI) provide authentication and authorization through digital signatures [2], [4]. Privacy-enhancing credential systems further support selective disclosure and unlinkability. However, both traditional and privacy-preserving trust infrastructures must address operational risks associated with credential misuse, authorization changes, consent withdrawal, and regulatory requirements. Consequently, revocation remains an essential mechanism despite its operational cost and the potentially irreversible effects of invalidation.[*1]

Timestamp-based revocation provides freshness guarantees and compatibility with existing PKI deployments, whereas accumulator-based revocation enables compact and efficient verification [8]. While these approaches are often viewed as alternatives, low-latency critical systems require both local verifiability and global consistency, motivating an integrated approach.

Technological progress in computing brings further operational risk with it to the trust infrastructure, which Post-Quantum Cryptography (PQC) addresses in providing resistance against known quantum attacks, including Shor's algorithm against public-key cryptography and Grover-type attacks against symmetric cryptography [13]. Representative examples include ML-KEM, ML-DSA, AES-256, SHA-256-based constructions, and ChaCha20-Poly1305 with appropriate security parameters. Although these mechanisms are currently regarded as secure against known attacks, their security remains computational in nature.

[1] Independent Researcher, Fujisawa, Kanagawa, 251–0032, Japan
[2] HISAFE, Ltd., Ebina, Kanagawa, 243–0405, Japan
[3] Shonan Institute of Technology, Fujisawa, Kanagawa, 251–8511, Japan
[4] Kansai University, Suita, Osaka, 564–8680, Japan
[a)] swohlge@gmail.com
[b)] kazuo.takaragi.wg@gmail.com
[c)] umezawa@info.shonan-it.ac.jp
[d)] gakuchohisho@ml.kandai.jp

[*1] Revocation is not claimed to be a complete legal solution to the "right to be forgotten". Rather, it is a technical enforcement component that supports data minimization, consent withdrawal, access termination, and cryptographic erasure within a well-defined governance framework.

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This limitation becomes increasingly important in systems where encrypted information, authentication records, transaction histories, operational logs, or digital evidence may retain value for extended periods. A discussion of the interpretation of Grover-type quantum search and its implications for long-term confidentiality is provided in Appendix A.1.

Long-term confidentiality is further challenged by harvesting attacks ("harvest (collect) now, decrypt later"), in which adversaries collect encrypted information today and defer cryptanalysis until future computational or algorithmic advances make recovery feasible [7]. Recent studies indicate that large-scale collection of encrypted communications can be economically inexpensive, shifting concern from data acquisition to future decryption capability [1]. Consequently, risks associated with sensitive information may accumulate throughout its lifetime even when immediate decryption is infeasible.

Economic models of information sharing capture incentives and risk [18], but are rarely unified with cryptographic trust mechanisms or extended to delayed risk realization. As a result, existing approaches do not fully connect authentication, privacy protection, revocation, long-term confidentiality, and their economic consequences within low-latency critical systems.

This paper addresses the problem of achieving secure, privacy-preserving, and economically interpretable digital trust management for low-latency critical systems operating under long-term cyber and cryptographic risks. Rather than strengthening individual cryptographic mechanisms in isolation, we establish a unified framework that jointly considers authentication, revocation, privacy protection, long-term confidentiality, and operational risk.

Digital identity usage is therefore treated as an economic activity whose risks, costs, and benefits evolve over time.

The main contributions of this paper are as follows:

- an economic trust model that unifies Type I and Type II errors and links them to measurable loss,
- an extension of this model to time-delayed risks, including harvesting attacks,
- a unified digital trust architecture (SK4SC: Secure Kernel for Supply Chains) to control sessions and transactions, respectively, that integrates post-quantum trust mechanisms, hybrid revocation, and accountable evidence management, and
- an information-theoretic extension based on IK techniques that complements computational security assumptions for long-term confidentiality.

The remainder of this paper is organized as follows. Section 2 introduces the economic trust model and formalizes the trade-off between Type I and Type II errors. Section 3 extends this model to incorporate harvesting attacks and delayed loss realization. Section 4 presents the SK4SC framework as a unified digital trust layer for low-latency critical systems. Section 5 discusses integration with distributed ledgers. Section 6 analyzes security and performance. Finally, Section 7 concludes the paper.

2. Economic Trust Model

Open information economy refers to an environment in which data and digital services can be shared freely or with minimal restrictions across organizational boundaries under verifiable trust and accountability.

This section formalizes the economic perspective underlying digital trust in LLCS. The proposed model treats the use of digital identity as an economic action with measurable benefit, cost, and risk, thereby integrating cryptographic verification with operational risk management.

A central objective is to establish a framework in which Type I and Type II errors can be balanced in an economically consistent manner. This provides the first required technical breakthrough: a principled mechanism to jointly reason about usability loss and security risk.

2.1 Key Concepts

- Each authorized use of digital identity corresponds to a micro-transaction,
- privacy risk is quantified as an expected operational cost,
- digital trust is a collective term for mechanisms that ensure "trust" in transactions and procedures within a digital society. and
- audit records provide verifiable cryptographic evidence of identity usage, provenance, and resulting economic outcomes.

Such privacy evidence supports accountability and can help mitigate the economic impact of erroneous decisions by enabling later review, dispute resolution, and recovery actions.

2.2 Formal Model of Utility and Loss

We model each credential usage event as a decision with expected utility. Let:

- a: action
- $v(a)$: benefit
- $c(a)$: cost
- $p_I(a)$: Type I error
- $p_{II}(a)$: Type II error
- L_I, L_{II}: losses

$$\mathcal{L}(a) = p_I(a)L_I + p_{II}(a)L_{II} \tag{1}$$

$$\mathcal{U}(a) = v(a) - c(a) - \mathcal{L}(a) \tag{2}$$

Decision rule:

$$\mathcal{U}(a) \geq 0 \tag{3}$$

2.3 Economic Interpretation of Error Trade-off

- Type I: usability loss as availability loss of information caused by rejection of a valid entity
- Type II: security loss caused by acceptance of an invalid entity

$$\min(p_I(a)L_I + p_{II}(a)L_{II}) = \min \mathcal{L}(a) \tag{4}$$

This defines trust as an economically optimized trade-off.

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2.4 Role of Revocation and Audit

- Accumulator: reduce Type I error
- Timestamp: reduce Type II error
- Audit: mitigate the economic consequences of Type I error and provide accountability for Type II error

$$p_I = f_I(\text{revocation}, \text{audit}), \quad p_{II} = f_{II}(\text{revocation}, \text{freshness}) \tag{5}$$

Here, audit does not directly prevent Type I or Type II errors. Instead, audit evidence records the provenance and outcomes of identity usage, enabling post-event verification, dispute resolution, recovery actions, and loss attribution. In the case of Type I errors, audit records may help recover otherwise lost opportunities or justify compensation. In the case of Type II errors, audit records help identify misuse, quantify resulting losses, and support accountability.

2.5 Limitation: Time-Blind Risk Model

The above framework evaluates risk at decision time. However, it fails to capture delayed loss caused by future cryptographic compromise.

> **"Security decisions that are optimal today may become economically catastrophic in the future."**

This motivates the need for explicit modeling of time-dependent risk.

3. Economic Loss Realization under Harvesting Attacks and Its Relation to SK4SC Verifiable Privacy Evidence

This section extends the economic model by incorporating harvesting attacks ("harvest (collect) now, decrypt later") and demonstrating the necessity of a second technical breakthrough: protection against future decryption based on information-theoretic security.

3.1 Probabilistic Model of Cryptographic Breakthrough (Q-Day)

We model the breakthrough time as a random variable:

$$T_Q \sim f_Q(t) \tag{6}$$

The precise form of $f_Q(t)$ is not assumed in this paper. Figure 1 is an illustrative distribution introduced solely to represent uncertainty in the arrival time of a cryptographically relevant quantum computer (CRQC). This modeling choice is consistent with recent quantum-threat timeline studies, which characterize Q-Day as a probability distribution rather than a deterministic date [12].

Rather than a fixed event, cryptographic compromise occurs with uncertainty over time.

Figure 1 highlights a critical insight that the breakthrough is uncertain and may occur earlier than expected, implying that risk cannot be postponed safely.

3.2 Action Model under Normal Operation

The SK4SC layer provides cryptographically verifiable

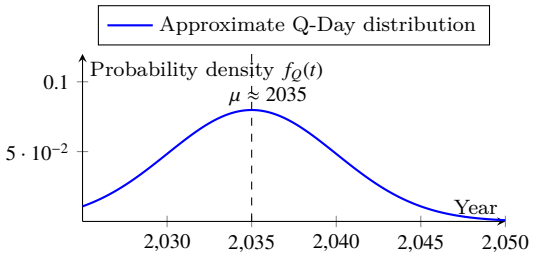


Fig. 1 Approximate Q-Day distribution with wide uncertainty and long tail.

privacy evidence for each action a:

- authentication and authorization proofs,
- revocation status (freshness and accumulator-based),
- audit and provenance records, and
- delegation structure and usage history.

These inputs define a structured evidence set:

$$E(a) = \{E_{\text{auth}}, E_{\text{rev}}, E_{\text{audit}}, E_{\text{deleg}}\} \tag{7}$$

which parameterizes the economic model by determining:

$$p_I(a), \quad p_{II}(a), \quad c(a) \tag{8}$$

If

$$p_{II}(a_t) \ll 1 \tag{9}$$

then the system appears secure.

3.3 Harvesting and Delayed Loss

Adversary accumulates data:

$$D = \{C_1, \ldots, C_T\} \tag{10}$$

After T_Q:

$$p_{II}(a_\tau) \to 1 \tag{11}$$

This transforms safe actions into compromised ones.

3.4 Long-term decision-making models in trust management

Traditional trust management focuses on verifying whether actions satisfy distributed authorization policies and trust credentials [2]. In contrast, the trust-management model considered in this paper extends the scope toward economic and operational decision-making under uncertainty, including long-term cyber, regulatory, and socio-economic effects.

The objective function is extended to incorporate uncertainty in the time of cryptographic compromise (Q-Day):

$$\min_{\{a_t\}} \mathbb{E}_{T_Q}\left[\mathbb{E}\left[\sum_t \gamma^t L_t(a_t \mid T_Q)\right]\right] \tag{12}$$

Equivalently,

$$= \min_{\{a_t\}} \int_0^\infty \left(\mathbb{E}\left[\sum_t \gamma^t L_t(a_t \mid T_Q = \tau)\right]\right) f_Q(\tau)\, d\tau \tag{13}$$

where $\gamma \in (0, 1]$ is a discount factor that weights future

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losses relative to immediate losses. Smaller values emphasize short-term considerations, whereas values close to one place greater weight on long-term consequences.

The minimization is taken over the sequence of actions $\{a_t\}$, including authentication strictness, revocation strategy, audit usage, data-retention policy, and the adoption of long-term protection mechanisms. Consequently, Eq. 13 optimizes not a single decision but an entire policy over time.

From a trust-management perspective, the environment in which decisions are made evolves continuously. Economic outcomes are influenced by evolving cyber, market, regulatory, and societal conditions, modeled here as stochastic disturbances affecting future losses. These influences can be represented as stochastic disturbances affecting the future loss process L_t.

The formulation is closely related to dynamic programming and stochastic control approaches widely used in engineering systems and economic decision theory. Unlike conventional trust management frameworks that focus primarily on verifying actions against security policies [2], the proposed model evaluates trust as a dynamic decision problem in which present actions influence future economic exposure under uncertain evolution of both cyber and socio-economic environments. In this sense, trust management complements risk management by supporting confidence-based decision making under uncertainty rather than merely reducing immediate risks.

The key insight is that even if the system appears secure at present, i.e., $p_{II}(a_t) \ll 1$, the expectation over T_Q captures the possibility that information judged acceptable today may generate substantial losses in the future. Thus, low-probability but high-impact events contribute to current expected loss.

Closed-form evaluation of Eq. 13 is generally difficult because policy decisions interact with uncertainty in both T_Q and the evolving environment. In practice, the expectation can be approximated using Monte Carlo simulation or scenario-based forecasting. Samples of T_Q and environment variables are drawn, system evolution is simulated under each scenario, and discounted losses are averaged.

A more detailed discussion is given in Appendix A.5.

3.5 Implication for SK4SC

SK4SC evidence guarantees present-time validity, but does not eliminate future compromise:

$$\mathbb{E}_{T_Q}[p_{II}(a_t \mid T_Q)] \gg p_{II}(a_t) \tag{14}$$

These observations imply that present-time verification alone is insufficient, as future compromise risk remains. This highlights the limitation of purely computational security and motivates IK-based randomness-reconstruction mechanisms that are intended to provide information-theoretic confidentiality under the assumptions discussed in [15].

The IK-based approach reduces sensitivity to T_Q and mitigates retrospective decryption risk. In this paper, IK-based extensions refer to a communication framework in which multiple parties can utilize independent true random number generators and jointly perform multi-party computation (MPC) — including multiplicative operations — between sender and receiver. Through this process, cryptographic and authentication operations can be constructed such that certain components achieve information-theoretic security, rather than relying solely on computational hardness assumptions. A more detailed discussion is given in Appendix A.2 and A.4.

3.6 Requirement for Information-Theoretic Security

To address this,

- protection must not depend on computational hardness,
- confidentiality must survive future breakthroughs, and
- harvesting must become economically irrational.

This motivates IK-based extension.

4. SK4SC: Post-Quantum Digital Trust Layer

4.1 Positioning of SK4SC in Digital Trust

As discussed in Section 2, there exists a conceptual gap between (a) cybersecurity with its trust model of access control and zero-trust assumptions, and (b) trust management with its operational risk model and economic decision framework.

We position SK4SC as an intermediate abstraction layer bridging these domains. SK4SC provides *digital trust* by offering verifiable and privacy-preserving evidence on authentication, authorization, and compliance with delegated rights, following the paradigm of privacy-enhanced credentials and auditable delegation [4].

In particular, SK4SC supports

- verifiable evidence of authentication without revealing identity or delegation structure,
- accountability of secondary use of information,
- integration of economic incentives (price mechanism), and
- operational risk management based on cryptographic evidence.

4.2 SK4SC as Secure CBDC-based Identity Kernel

SK4SC is realized using the secure CBDC framework [16], which provides an identity and credential infrastructure for CPS and digital economy applications.

The core requirements are

(i) selective disclosure of attributes,
(ii) hiding of issuer delegation chains,
(iii) privacy-preserving revocation / non-revocation proofs,
(iv) compatibility with distributed ledger systems, and
(v) long-term post-quantum security.

Earlier designs relied on pairing-based Delegatable

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Anonymous Credentials (DAC) and zero-knowledge range proofs [3]. However, such constructions are not secure against quantum adversaries, motivating a redesign based on post-quantum cryptography [13], [14].

4.3 Unified Post-Quantum Construction

A core requirement of SK4SC is support for DAC, which enable credential holders to prove authorization rights and delegation relationships without revealing unnecessary identity information.

While earlier DAC constructions relied primarily on pairing-based cryptography, SK4SC adopts a post-quantum approach based on lattice cryptography.

The key design principle of SK4SC is that all verification conditions are expressed as a single NP statement over lattice-based relations, following the paradigm of lattice-based zero-knowledge proofs [9], [10]. Hence, a list of parameter definitions and some brief explanations have been added as follows:

$$\Pi = \mathrm{NIZK}\left\{ \exists s, r, a, d : \begin{array}{l} A_{\mathrm{chain}} s = t_{\mathrm{chain}} \pmod q, \\ C_i = A r_i + a_i G, \\ L_i \le a_i \le U_i, \\ \mathrm{RevocationCondition}(d) = 1, \\ \mathbf{s}\backslash, \mathbf{r}\backslash \le \beta \end{array} \right\} \tag{15}$$

This unified structure simultaneously ensures

- DAC delegation-chain validity (without revealing intermediate issuers),
- attribute commitment consistency and selective disclosure,
- range constraints on sensitive attributes, and
- revocation / non-revocation proof correctness.

Thus, SK4SC replaces signature composition (pairing-based approaches) with proofs of existence of short witnesses, which is the standard paradigm in lattice-based cryptography.

4.4 Revocation Mechanisms (Unified View)

In LLCS, revocation is challenging because verification must often be performed locally under intermittent connectivity and potentially stale revocation information.

In the proposed framework, revocation is viewed not only as a cryptographic validity check but also as a mechanism for managing the economic trade-off between Type I (false rejection) and Type II (false acceptance) errors. Timestamp-based non-revocation evidence and accumulator-based revocation represent different operating points in this trade-off space and can be combined according to operational requirements.

4.4.1 Timestamp-Based Non-Revocation Evidence (Type II Error Minimization)

This approach prioritizes the prevention of Type II errors, i.e., accepting a credential that should no longer be trusted because it has been revoked, invalidated, or subsequently associated with policy violations.

A verifier requires authoritative evidence that a credential was valid at a specific point in time. This reduces the probability of accepting untrustworthy credentials at the cost of additional verification overhead.

Economically, it trades higher verification cost for lower security risk and is therefore suitable when the potential loss from a Type II error is high.

4.4.2 Accumulator-Based Revocation (Type I Error Minimization)

This approach prioritizes the reduction of Type I errors, i.e., unnecessarily rejecting valid entities.

Verification is compact and can be performed locally using cached accumulator information, resulting in low latency and high scalability. The trade-off is that stale revocation information may temporarily increase exposure to Type II errors.

Accordingly, accumulator-based revocation is well suited to high-frequency interactions where operational efficiency is a primary concern.

4.4.3 Hybrid Revocation Strategy as Economic Control

The key contribution of SK4SC is not a new revocation algorithm, but a unified interpretation of revocation as a dynamic control mechanism over the Type I / Type II trade-off:

- accumulator-based revocation favors operational efficiency and lower Type I error, while
- timestamp-based validation favors stronger trust assurance and lower Type II error.

The system can adaptively select between these mechanisms according to transaction value, operational context, expected benefit, and risk tolerance.

The objective is not to minimize revocation itself, but to maximize expected utility by balancing expected benefit against verification cost and security risk. Routine transactions may use accumulator-based validation, whereas high-value or policy-sensitive transactions may require timestamp-based evidence.

Consequently, revocation becomes more than a binary validity decision. It acts as an economic control layer that adjusts the balance among usability, cost, and security under uncertainty.

In this sense, revocation serves as a pricing mechanism of trust: higher trust assurance generally requires additional validation cost, while lower assurance may increase exposure to future losses. Revocation policy therefore internalizes trust-related risks into economic decision making, enabling risk-aware operation in autonomous decentralized CPS.

Although revocation cannot prevent all future misuse of previously disclosed confidential information, it supports authorization termination, accountability, and limitation of continued use after a trust relationship has changed. This is also relevant to governance objectives such as consent withdrawal, cryptographic erasure, and practical aspects of the “right to be forgotten”.

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5. Integration with Distributed Ledger

In lattice-based anonymous credential systems, multi-attribute credential presentations may reach sizes on the order of tens of kilobytes. If stored directly on-chain, this leads to

1) reduced transaction throughput per block,
2) increased endorsement latency,
3) storage overhead, and
4) higher peer-to-peer resynchronization cost.

5.1 Recommended Architecture

To address these challenges, we adopt the following architecture:

1) **On-ledger:** record only commitments, hash references, attestations, and compact revocation summaries.
2) **Off-ledger:** store large zero-knowledge proofs, certificates, and witnesses in P2P or content-addressable storage.
3) **Batch verification:** support collective verification of multiple proofs.
4) **Aggregation:** aggregate signatures and attributes within epochs to reduce updates.
5) **Epoch design:** apply revocation updates in batches to balance responsiveness and efficiency.

5.2 Notes on Description

In post-quantum settings, it is preferable to describe the integration as:

> **"Proofs are verified off-ledger, while commitments and attestations are anchored on-ledger."**

This formulation reflects the increased proof size and aligns with scalable credential systems [4].

5.3 SK4SC Overview Diagrams

Figure 2 and Figure 3 illustrate the role and operation of SK4SC.

5.3.1 System Structure

SK4SC functions as a digital trust layer bridging two domains (Fig. 2):

- cybersecurity, which enforces access control under zero-trust assumptions, and
- trust management, which evaluates risk, incentives, and economic outcomes.

SK4SC provides

- privacy-preserving authentication with selective disclosure,
- hybrid revocation for scalable and reliable validation,
- ledger-based auditability and provenance, and
- integration of post-quantum cryptography with IK-based extensions.

The key insight is that trust is achieved by linking cryptographic evidence with economic decision making in the open information economy, rather than relying on static trust assumptions.

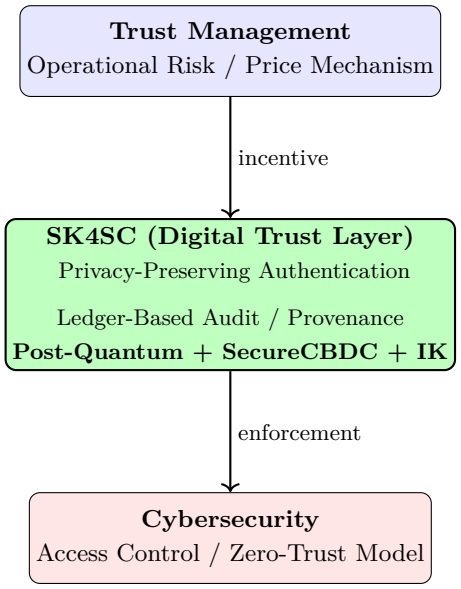


Fig. 2 SK4SC as a post-quantum digital trust layer bridging trust management and cybersecurity.

5.3.2 Data Flow

The system operates through two coupled flows: a cryptographic verification flow and an economic accountability flow (Fig. 3).

(A) Cryptographic verification flow:

- Entities present credentials as zero-knowledge proofs.
- SK4SC verifies correctness and policy compliance.
- Results are anchored on the ledger as commitments or attestations.
- Audit evidence supports subsequent trust evaluation.

(B) Economic accountability flow:

- Each use of digital identity is treated as a micro-transaction.
- Usage generates measurable benefit and cost.
- Value transfer or incentives are associated with usage events.
- Usage history is accumulated as audit-linked evidence.

These flows are connected through ledger-anchored data, enabling both correctness verification and economic accountability.

By using the same evidence base for authorization, auditability, risk evaluation, and economic decision making, SK4SC establishes a common framework that connects authentication outcomes with their operational and economic consequences. This allows trust to be managed not only as a security property but also as an economically accountable process for CPS and digital economy systems.

6. Security Model and Performance Considerations

6.1 Security Model and Goals

We consider an environment in which multiple autonomous entities exchange authenticated information under intermittent connectivity, delegated trust relationships, and evolving operational risks. The adversary is

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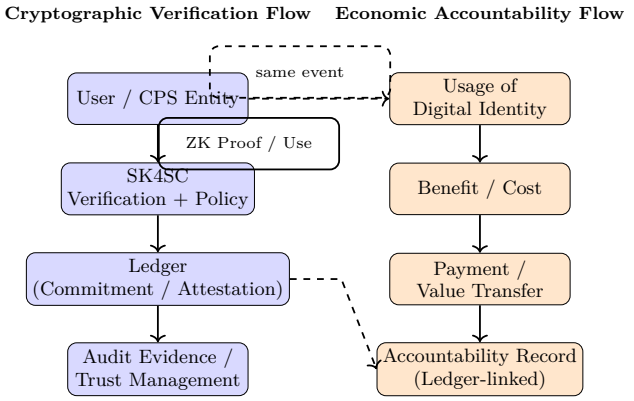


Fig. 3 Data flow in SK4SC: left (blue): cryptographic verification flow; right (orange): economic accountability flow associated with each use of digital identity.

assumed to control the communication network, observe protocol executions, store transmitted ciphertexts and credentials, and attempt future cryptanalysis on recorded data. The adversary may also exploit implementation weaknesses or system misconfigurations, potentially assisted by advanced AI-based analysis.

Under this model, the proposed framework aims to achieve the following security goals.

6.1.1 Authentication Soundness

An adversary should not be able to convince a verifier that it possesses a valid credential or delegated right unless it actually holds a corresponding valid witness. This property relies on the soundness of the underlying lattice-based proof system and the unforgeability of the associated post-quantum primitives.

6.1.2 Revocation Correctness and Freshness

In our framework, accumulator-based revocation provides efficient local verification using cached revocation state, while timestamp-based non-revocation evidence provides authoritative refresh of trust when local state becomes stale or policy requires stronger validation.

6.1.3 Privacy-Preserving Presentation

Credential presentation should reveal no more information than required for authorization. In particular, intermediate delegation structure, unnecessary attributes, and linkable identity information should remain hidden, except for what is intentionally disclosed by policy. This property is supported by zero-knowledge proof techniques and selective disclosure mechanisms.

6.1.4 Accountability and Auditability

Although identity information may be selectively hidden during presentation, authorized use must remain auditable through commitments, attestations, and ledger-anchored evidence. This enables later verification of compliance with delegated rights and supports operational risk management in the open information economy.

6.1.5 Long-Term Confidentiality

This goal is especially important for CPS-related information whose value persists over long periods, including operational logs, digital twin data, and sensitive personal information.

To capture this requirement, we distinguish two layers of confidentiality:

- **Computational confidentiality** provided by post-quantum cryptography under standard hardness assumptions against classical and quantum adversaries.
- **Information-theoretic confidentiality** strengthened by IK-based extensions, which aim to preserve secrecy even if computational assumptions used in the cryptographic layer are weakened or broken in the future.

6.1.6 Resistance to Harvesting Attacks

We explicitly consider adversaries that record encrypted traffic or stored ciphertexts today and attempt to decrypt them later. In such attacks, the time gap between data collection and cryptanalysis invalidates security models that consider only present-time adversaries. Our framework addresses this threat by combining post-quantum protection with IK-based extensions, thereby reducing the risk that future cryptanalytic breakthroughs will retroactively compromise past confidentiality.

6.1.7 AI-Assisted Adversaries

AI may accelerate exploitation of weaknesses and reduce practical security margins.

6.2 Security Interpretation of the Proposed Architecture

The architecture of SK4SC should be understood as a composition of complementary security layers.

First, the post-quantum credential and proof layer provides computational protection for authentication, selective disclosure, and revocation-related validation. This layer supports practical deployment and interoperability with modern cryptographic infrastructures.

Second, the hybrid revocation layer balances local autonomy and global consistency. Accumulator-based revocation allows efficient validation under disconnection or edge operation, whereas timestamp-based evidence restores stronger trust when local cache freshness is insufficient. Neither mechanism alone fully addresses the CPS requirement of autonomous decentralized operation with policy-compliant recovery; their combination is therefore essential.

Third, the ledger-anchored audit layer provides accountability, provenance, and consistency over time without placing heavy verification directly on the real-time control path.

6.3 Performance and Deployment Implications

The proposed security model has practical implications for deployment. Lattice-based post-quantum proofs increase proof size and verification cost compared with some classical constructions. Therefore, direct placement of full proofs on the real-time control path or on-chain is generally undesirable. In our architecture, bulky proofs and witnesses are processed off-ledger, while commitments and

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attestations are anchored on-ledger. See Appendix A.6 for performance evaluation and concrete instantiation.

For CPS deployments, the architecture should separate

- real-time control decisions,
- local authentication and cache-based revocation checks, and
- asynchronous ledger synchronization and audit processing.

This enables the system to preserve responsiveness in control loops while still maintaining cryptographic assurance and auditability. The use of hybrid revocation improves deployment flexibility by allowing fast local decisions during normal operation and stronger trust refresh when required by policy or critical operating conditions.

7. Conclusion

Economically controlled openness via unified error management and hybrid revocation: The proposed framework establishes an open information economy in which participation is governed by verifiable identity usage and economically consistent control of Type I (false rejection) and Type II (false acceptance) errors. A central contribution is the unified treatment of these error types within an economic model, together with a hybrid revocation strategy that combines accumulator-based mechanisms and timestamp-based non-revocation evidence. This complementary use enables context-dependent optimization of availability and security, demonstrating that revocation functions as a dynamic control mechanism of trust.

Resilient participation enabled by IK-based long-term security: The integration of IK-based extensions provides a new mechanism for mitigating time-delayed risks arising from harvesting attacks and future cryptographic compromise. In combination with post-quantum cryptography and adaptive revocation, this enables continuity of participation even under dynamic trust conditions and long-term uncertainty, representing a key added value.

Evidence-based risk-aware LLCS operation: In cyber-physical systems operating under uncertainty, the integration of cryptographic verification and audit-derived evidence enables decisions that are both secure and economically interpretable.

Additional systemic effects on society: Since the unified digital trust framework takes dynamic behavior over time into account to motivate secure and privacy-preserving use of information with third parties, informatics operates as a tool to justify trust assumptions in information sharing and so to discover new knowledge by scientific means. This layer is intended to incentivize the behavior on information exchange with accountability; and, it operates using cryptographically secured privacy evidence, adapted to the specific situation and timing.

It provides a consistent foundation for an open information economy in which trust is not assumed, but verifiably constructed, economically controlled, and sustained over time.

References

[1] J. Blanco-Romero, F. A. Mendoza, C. G. Rubio, C. Campo, and D. D. Sanchez, "On the practical feasibility of harvest-now, decrypt-later attacks," arXiv:2603.01091, 2026. https://doi.org/10.48550/arXiv.2603.01091

[2] M. Blaze, J. Feigenbaum, and J. Lacy, "Decentralized Trust Management," IEEE S&P 1996, IEEE, pp. 164–173, 1996. https://doi.org/10.1109/SECPRI.1996.502679

[3] B. Bünz, J. Bootle, D. Boneh, A. Poelstra, P. Wuille, and G. Maxwell, "Bulletproofs: Short Proofs for Confidential Transactions and More," IEEE S&P 2018, IEEE, pp. 315–334, 2018. https://doi.org/10.1109/SP.2018.00020

[4] J. Camenisch, M. Drijvers, and M. Dubovitskaya, "Practical UC-secure delegatable credentials with attributes and their application to blockchain," CCS 2017, ACM, pp. 683–699, 2017. https://doi.org/10.1145/3133956.3134025

[5] M. Christ, F. Baldimtsi, K. K. Chalkias, D. Maram, A. Roy, and J. Wang, "SoK: Zero-Knowledge Range Proofs," Cryptology ePrint Archive, Paper 2024/430, 2024. https://eprint.iacr.org/2024/430

[6] CRYSTALS-Dilithium Team, "CRYSTALS-Dilithium reference implementation," https://github.com/pq-crystals/dilithium

[7] F. Kagai, P. Branch, J. Blut, and R. Allen, "Harvest-Now, Decrypt-Later: A Temporal Cybersecurity Risk in the Quantum Transition," Telecom, 2025. https://doi.org/10.3390/telecom6040100

[8] V. Y. Kemmoe, A. Lysyanskaya, and N. K. Nguyen, "Lattice-Based Accumulator and Application to Anonymous Credential Revocation," Cryptology ePrint Archive, Paper 2025/1099, 2025. https://eprint.iacr.org/2025/1099

[9] V. Lyubashevsky, "Fiat-Shamir with Aborts: Applications to Lattice and Factoring-Based Signatures," ASIACRYPT 2009, LNCS, vol. 5912, Springer, pp. 598–616, 2009. https://doi.org/10.1007/978-3-642-10366-7_35

[10] V. Lyubashevsky, "Lattice Signatures without Trapdoors," EUROCRYPT 2012, LNCS, vol. 7237, Springer, pp. 738-755, 2012. https://doi.org/10.1007/978-3-642-29011-4_43

[11] R. C. Merkle, "A Digital Signature Based on a Conventional Encryption Function," CRYPTO '87, LNCS, vol. 293, pp. 369–378, Springer, 1988. https://doi.org/10.1007/3-540-48184-2_32

[12] M. Mosca and M. Piani, "Quantum Threat Timeline Research Report 2024," evolutionQ Publication, 2025. https://www.evolutionq.com/publications/quantum-threat-timeline-research-report-2024

[13] NIST, "Post-Quantum Cryptography Standards (FIPS 203–205)," 2024–. https://csrc.nist.gov/pqc-standardization

[14] NIST, "Module-Lattice Digital Signature Standard (ML-DSA)," FIPS 204, 2024. https://doi.org/10.6028/NIST.FIPS.204

[15] K. Takaragi, K. Furukawa, K. Umezawa, and S. Wohlgemuth, "Low-latency critical communication infrastructure with long-term secrecy," CSS 2026, Hamamatsu, 2026.

[16] K. Takaragi, T. Kubota, S. Wohlgemuth, K. Umezawa, and H. Koyanagi, "Secure Revocation Features in eKYC - Privacy Protection in Central Bank Digital Currency," IEICE Trans. Fundamentals, vol. E106-A, no. 3, pp. 325–332, 2023. https://doi.org/10.1587/transfun.2022CIP0008

[17] M. N. Wegman and J. L. Carter, "New Hash Functions and Their Use in Authentication and Set Equality," Journal of Computer and System Sciences, vol. 22, no. 3, pp. 265–279, 1981. https://doi.org/10.1016/0022-0000(81)90033-7

[18] S. Wohlgemuth, K. Takaragi, T. Kubota, K. Umezawa, and K. Hasegawa, "Assuring Trust in Confidential Data Use by Identity Management," SCIS 2024, Nagasaki, Japan, 2024.

Appendix

A.1 Interpretation of Grover-Type Quantum Search for Symmetric Cryptography

This appendix briefly clarifies the security implications of Grover-type quantum search for symmetric cryptography and explains why long-term confidentiality remains a

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concern even in post-quantum settings.

A common misunderstanding is that the statement

> "an n-bit symmetric primitive provides approximately $n/2$ bits of post-quantum security"

means that a quantum computer can directly derive part of the secret key in the same manner that Shor's algorithm breaks RSA or elliptic-curve cryptography. This interpretation is incorrect.

A.1.1 Difference from Shor's Algorithm

Shor's algorithm exploits algebraic structure in public-key cryptographic problems such as integer factorization and discrete logarithms. In principle, it can recover secret keys from publicly available information.

In contrast, Grover's algorithm does not exploit a mathematical weakness of the underlying cipher. Instead, it provides a quadratic speedup for unstructured search. When applied to symmetric cryptography, it reduces the cost of exhaustive key search from approximately 2^n operations to approximately $2^{n/2}$ operations.

Therefore, Grover search should be understood as an acceleration of brute-force search rather than a direct cryptanalytic break of the cryptographic primitive.

A.1.2 Practical Interpretation

For currently deployed symmetric cryptography, no generally accepted quantum algorithm is known that provides a larger advantage than the quadratic speedup represented by Grover search.

Consequently, AES-256 is commonly regarded as providing roughly a 128-bit security level against currently known quantum attacks. This does not imply that practical key recovery is easy, since large-scale attacks would require fault-tolerant quantum computers capable of performing a very large number of coherent cipher evaluations.

However, these estimates are based on currently known algorithms and current understanding of quantum computing. They should not be interpreted as unconditional guarantees against all future advances in cryptanalysis, quantum algorithms, or computing architectures.

A.1.3 Implications for Long-Term Confidentiality

AES-128, AES-256, and related symmetric cryptographic mechanisms are not currently considered broken by quantum computers. Nevertheless, systems that must protect information for decades face a broader challenge.

An adversary may collect encrypted information today and retain it for future analysis. Although current post-quantum security estimates suggest strong protection, future advances in algorithms or computational capabilities could alter present assumptions regarding the cost of cryptanalysis.

For this reason, long-term security cannot rely solely on contemporary computational assumptions. Larger security margins such as AES-256 remain important, but they are still computational in nature.

This observation motivates the use of information-theoretic extensions such as IK-based mechanisms. While post-quantum cryptography provides protection against currently known quantum attacks, information-theoretic techniques can further reduce dependence on future computational assumptions and help mitigate retrospective decryption risks associated with long-lived data.

A.2 Long-term Security and IK Extension

Long-term confidentiality is critical for CPS infrastructures, where data may retain value over extended periods. While post-quantum cryptography provides resistance against known quantum attacks, it remains based on computational hardness assumptions.

To address this limitation, we incorporate information-theoretic IK-based randomness reconstruction mechanisms. For details on IK extensions see [15] and its variation IK-light in A.4. Unlike computational cryptography, IK-based mechanisms do not rely on assumptions about adversarial computational capabilities or future advances in attack techniques.

In the proposed architecture, IK extensions complement post-quantum cryptography by

- reducing dependence on long-term computational security assumptions,
- preventing retrospective decryption of stored ciphertexts,
- mitigating risks from future cryptanalytic and algorithmic advances, and
- providing an independent layer of long-term protection.

Thus, the framework achieves a dual-layer security model: The computational layer supports practical deployment, while the information-theoretic layer strengthens long-term confidentiality against both foreseeable and unforeseen future breakthroughs.

A.3 Lagrange Interpolation and MPC-Based Secure Computation

This appendix briefly reviews Lagrange interpolation and its role in secure Multi-Party Computation (MPC). MPC enables multiple organizations to jointly process sensitive information without revealing their underlying data. Typical applications include healthcare, insurance, financial services, and inter-organizational analytics.

A widely used MPC technique is Shamir secret sharing over a finite field $\mathbb{F}_q$. A secret

$$s \in \mathbb{F}_q \tag{A.1}$$

is embedded as the constant term of a random polynomial

$$f(x) = s + w_1 x + \cdots + w_{t-1} x^{t-1}. \tag{A.2}$$

Each participant receives a share $f(x_j)$ at a public nonzero point x_j. When at least t shares are available, the secret

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can be reconstructed using Lagrange interpolation:

$$s = f(0) = \sum_{j=1}^{t} f(x_j)\lambda_j \tag{A.3}$$

, where

$$\lambda_j = \prod_{\ell \neq j} \frac{x_\ell}{x_\ell - x_j} \tag{A.4}$$

The key property is that arithmetic operations can be performed directly on secret shares. Thus,

$$z = F(s_1, s_2, \ldots, s_n) \tag{A.5}$$

can be evaluated without revealing the private inputs, while only the authorized output is reconstructed. From an engineering perspective, an MPC system consists of

- distributed confidential databases,
- MPC processing nodes,
- secret-sharing mechanisms, and
- controlled reconstruction of results.

Sensitive information remains distributed throughout the computation, while useful results can be derived from the reconstructed output.

While MPC provides information-theoretic confidentiality for computations on secret shares, practical deployments often employ additional mechanisms such as authentication, audit logging, and digital signatures. The IK framework extends the MPC-based architecture by organizing these functions while preserving long-term confidentiality under defined trust assumptions. IK-light, introduced in the next section, represents a lightweight communication-oriented realization of this framework.

A.4 IK-light

The IK framework is not intended to expand a short secret into an information-theoretically secure long key stream. Instead, it enables the generation, verification, distribution, and reconstruction of true-random masks through distributed True Random Number Generators (TRNGs), secret sharing, and MPC.

This section introduces IK-light, a lightweight operational variant of the IK framework presented in [15]. While the original IK construction employs MPC-based processing to generate cryptographically protected communication resources, IK-light moves most computationally expensive operations into an offline preparation phase in order to minimize online latency.

Let the sender and receiver share a secret key k in advance. During the offline phase, random values generated through the IK framework are combined to obtain a random mask sequence

$$R = R_1 \| R_2 \| \cdots \tag{A.6}$$

, where each mask block R_i has the same length as the corresponding message block a_i. The shared key k is used only for authentication and authorization purposes. The randomness masks R_i are derived from independently generated true random values within the IK framework and are not generated from k.

The detailed generation procedure follows the IK construction described in [15]. For the present discussion, the important point is that the expensive MPC-related preparation is completed before communication begins.

During the online phase, encryption reduces to a simple XOR operation:

$$C_i = a_i \oplus R_i \tag{A.7}$$

The sender transmits C_i.

The receiver computes

$$\hat{a}_i = C_i \oplus R_i \tag{A.8}$$

Since XOR is its own inverse,

$$\hat{a}_i = (a_i \oplus R_i) \oplus R_i = a_i \tag{A.9}$$

Consequently, online encryption and decryption consist primarily of byte-wise XOR processing.

Authenticity and integrity are provided independently through a Wegman–Carter type Message Authentication Code (MAC).

Wegman and Carter showed that information-theoretically secure authentication can be achieved using universal hash functions combined with one-time randomness [17].

Let m denote a message. A typical Wegman–Carter MAC is expressed as

$$\mathrm{MAC}(m, k, r) = h(m, k) + b(r) \tag{A.10}$$

, where

- k is a secret authentication key,
- $h(m, k)$ is a universal hash function,
- r is a one-time random value, and
- $b(r)$ is a one-time masking value derived from r.

A representative universal hash function is polynomial hashing:

$$h(m, k) = \sum_{i=0}^{n-1} m_i k^i \tag{A.11}$$

, where

$$m = (m_0, m_1, \ldots, m_{n-1}) \tag{A.12}$$

Authentication succeeds only when the receiver recomputes the same hash value and obtains the identical authentication tag.

The engineering advantage of this architecture is that confidentiality and authenticity are implemented independently. Confidentiality is provided by IK-generated random masks, while authenticity and integrity are guaranteed by a Wegman–Carter authentication layer.

Modern embedded platforms increasingly include low-cost TRNG devices, TPM-class security modules, and dedicated hardware random-number generators. Such hardware makes practical deployment of one-time randomness increasingly feasible even in large-scale CPS environments.

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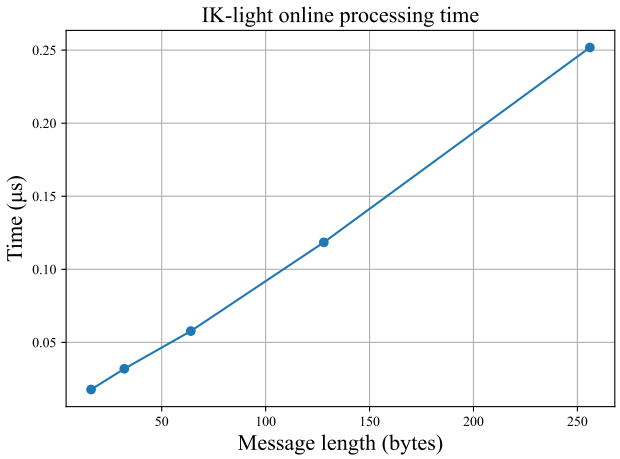


Fig. A·1 Online processing time of IK-light. The online processing consists of XOR-based encryption, XOR-based decryption, Wegman–Carter MAC generation, and MAC verification. The random masks R_i and one-time authentication randomness are assumed to be prepared offline, so their generation cost is not included in the online processing time.

The resulting architecture can be viewed as a two-layer information-theoretically secure design:

- IK-generated XOR masking for confidentiality, and
- Wegman–Carter authentication for authenticity and integrity.

Because both layers rely on fresh randomness, security depends strongly on proper randomness management. Reuse of a mask R_i introduces risks similar to one-time-pad reuse, while reuse of MAC randomness r degrades authentication security.

The primary motivation of IK-light is performance optimization. Once masks and authentication randomness have been prepared offline, online processing is dominated by

(1) XOR-based encryption,
(2) XOR-based decryption,
(3) Wegman–Carter MAC generation, and
(4) MAC verification.

As reported in [15], this substantially reduces online processing cost compared with the original IK-based implementation. See Figure A·1.

Therefore, IK-light may be regarded as a practical communication-oriented realization of MPC-generated cryptographic resources. It preserves the information-theoretic security philosophy of the original framework while achieving significantly lower online latency for high-throughput CPS communication systems.

A.5 Conceptual Visualization of Economic Shock

Figure A·2 is an illustrative conceptual sketch based on the qualitative behavior implied by Eq. 13. It is not derived from an empirical dataset and should not be interpreted as a prediction of actual utility values or breakthrough timing. Instead, it visualizes a representative scenario in which a cryptographic breakthrough (Q-Day) is realized at time t^*.

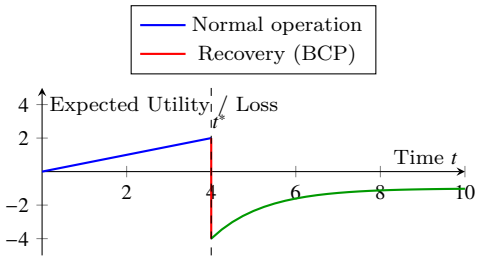


Fig. A·2 Impact of harvesting attack on expected utility over time.

Figure A·2 provides a strong intuitive message:

- utility grows under normal operation,
- a stochastic breakthrough triggers abrupt collapse, and
- recovery is costly and incomplete.

"Even without knowing when the attack succeeds, its probability already creates economic loss today."

A.6 Performance Evaluation and Concrete Instantiation

This appendix gives a compact instantiation of the proposed lattice-based SecureCBDC construction, with the purpose of clarifying the performance implications discussed in Section 6.3.

The numerical values shown in the figures of this appendix should be understood as illustrative estimates derived from the constraint-count model described below. They are intended to indicate relative scaling trends and deployment implications, rather than to report full implementation benchmarks.

A.6.1 Choice of ML-DSA

As a representative post-quantum signature primitive, we instantiate the issuer signatures with ML-DSA, standardized in NIST FIPS 204. ML-DSA is a post-quantum computational security mechanism. Its security relies on the assumed hardness of certain lattice problems, even for quantum adversaries. It is not information-theoretically secure. We use ML-DSA-65 as a reference parameter set, since it offers a balanced security and performance profile for applications requiring long-term authenticity.

The relevant operational point is that ML-DSA signatures and public keys are larger than classical elliptic-curve signatures. A typical ML-DSA-65 public key is about 1.9 KB and a signature is about 3.3 KB, as specified in FIPS 204 [14]. For the closely corresponding CRYSTALS-Dilithium3 parameter set, the official optimized AVX2 implementation reports signing and verification costs of about 5.3×10^5 and1.8×10^5 CPU cycles, respectively, on an Intel Core i7-class processor [6].

Thus, the basic signature layer is not expected to dominate the end-to-end cost of the proposed Delegatable Anonymous Credential (DAC) proof system; the larger

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contribution comes from embedding delegation, attribute, and revocation conditions into a single lattice-based zero-knowledge proof.

A.6.2 DAC Hierarchy

Figure A·3 illustrates the delegation hierarchy considered in the evaluation. The examples range from a two-level chain to a four-level chain:

$$\text{Root CA} \rightarrow \text{User} \tag{A.13}$$

$$\text{Root CA} \rightarrow \text{Local CA} \rightarrow \text{User} \tag{A.14}$$

, and

$$\text{Root CA} \rightarrow \text{Federation CA} \rightarrow \text{Local CA} \rightarrow \text{User} \tag{A.15}$$

Each issuer I_j holds an ML-DSA key pair (sk_j, pk_j), and delegation is represented as a signed public-key link. The user proves, without revealing the full delegation chain, that there exists a valid sequence of issuer signatures and attributes satisfying a single NP relation:

$$\exists\{s_j\} : \quad A_{\text{chain}} s_{\text{chain}} = t_{\text{chain}} \pmod{q} \tag{A.16}$$

This relation combines signature validity, delegation consistency, attribute conditions, and revocation status into one zero-knowledge statement.

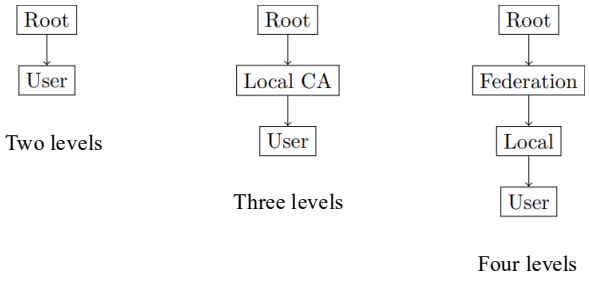


Fig. A·3 Example DAC delegation hierarchy with two to four levels.

A.6.3 Evaluation Methodology

The estimates in this appendix are not full implementation benchmarks. They are order-of-magnitude estimates obtained by decomposing the proof into its main lattice operations:

- matrix-vector products As,
- Fiat–Shamir hashing,
- verification of shortness constraints,
- range proofs for attributes or timestamp differences, and
- Membership or non-membership checks for revocation accumulators.

The number of constraints in the resulting NP relation is then used as a proxy for proof generation time, verification time, and proof size. This methodology is intended to compare design alternatives and scalability trends, rather than to claim exact implementation timings. For this reason, the plotted values in Figures A·4, A·5, and A·6 should be read as model-based estimates. They are calibrated to the scale of lattice signature and proof operations, but they should not be interpreted as measurements of a particular implementation.

A.6.4 Signature and Delegation Verification Cost

Figure A·4 shows the qualitative behavior of signing and verification cost as the delegation depth increases from two to four levels. Each additional delegation link contributes one additional issuer signature and one additional consistency constraint. Therefore, to first order,

$$C_{\text{prove}}, C_{\text{verify}} = O(k) \tag{A.17}$$

where k is the number of delegation levels. The important practical implication is that the cost grows approximately linearly with the delegation depth. Verification remains lighter than proof generation because it mainly recomputes challenges, checks linear relations, and verifies bounded-norm constraints. For the short delegation chains expected in SecureCBDC operation, this overhead is acceptable: the verifier does not need to perform an exhaustive search over issuers, nor does it learn the hidden chain.

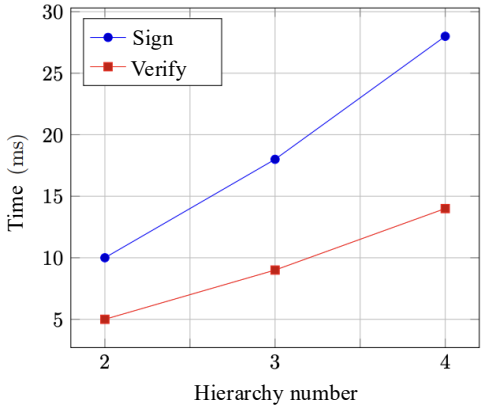


Fig. A·4 Illustrative signing and verification cost for ML-DSA-65-based DACs. The plotted values are model-based estimates derived from delegation depth and constraint-count scaling, not implementation benchmarks.

A.6.5 Revocation Proof

We compare two revocation mechanisms: a timestamp-comparison method and accumulator-based method. Figure A·5 gives the qualitative processing-time comparison, and Figure A·6 gives the corresponding proof-size comparison.

In the timestamp method, non-revocation or revocation is reduced to a zero-knowledge range proof over a timestamp difference. This follows the standard use of range proofs for showing that a committed or hidden value

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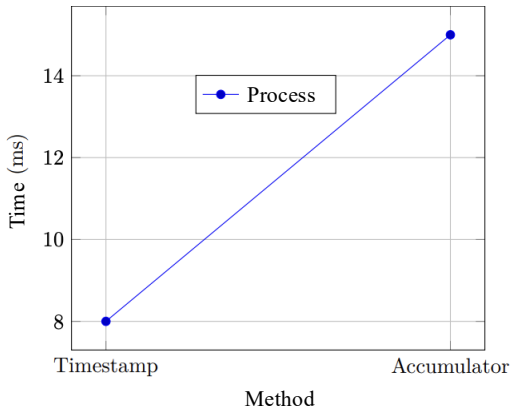


Fig. A·5 Illustrative processing-time comparison of revocation proof methods. The values are model-based estimates derived from the constraint-count approximation described in the text, rather than implementation benchmarks.

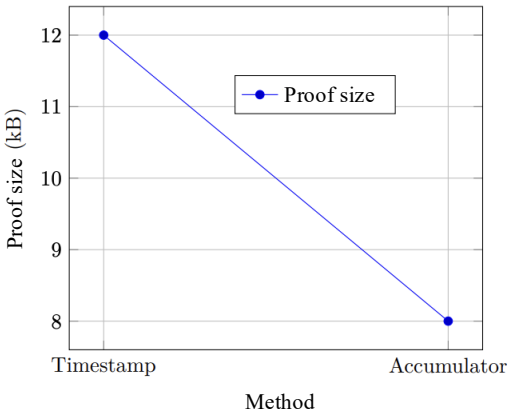


Fig. A·6 Illustrative proof-size comparison of revocation proof methods. The values indicate relative scaling trends under the model described in the text, not measured proof sizes of a particular implementation.

lies within a prescribed interval [5]. For example, non-revocation can be written as

$$d = t_{nr} - t_{sig}, \qquad 0 \le d \le D_{\max} \tag{A.18}$$

, where $D_{\max}$ is the maximum permitted time interval. Using bit decomposition,

$$d = \sum_{i=0}^{\ell-1} b_i 2^i, \qquad b_i \in \{0,1\}, \qquad \ell = \lceil \log_2 D_{\max} \rceil \tag{A.19}$$

the number of range-proof constraints grows as

$$O(\ell) = O(\log D_{\max}) \tag{A.20}$$

This method is attractive for low-latency use cases because the constraint structure is simple. In the present appendix, the timestamp values plotted in Figures A·5 and A·6 are therefore obtained by applying this logarithmic range-proof model to the timestamp difference, rather than by quoting a measured timestamp-authority or implementation-specific benchmark. In the accumulator method, the public accumulator value can remain compact even when the revocation set R grows.

However, the proof must include a witness-verification relation for membership or non-membership, written abstractly as

$$\mathrm{VerifyAcc}(Acc, pk, w) = 1 \tag{A.21}$$

When implemented using a tree-like witness path, as in Merkle-tree-based authenticated structures and lattice-based accumulator constructions, the proof component grows approximately as

$$O(\log |R|) \tag{A.22}$$

Such logarithmic dependence reflects the length of the authentication or witness path in tree-structured accumulators and related constructions[8], [11]. For example, if$|R| \approx 10^6 \approx 2^{20}$, the witness path has about 20 levels. The asymptotic growth is therefore logarithmic, but each level may involve more expensive algebraic or hashing constraints than a simple range proof.

The timestamp method is preferable when each proof is generated once and verified only a small number of times.

The accumulator method is preferable when revocation information is checked repeatedly or at high frequency, because verification can be performed locally as long as the accumulator state and witness are available. Its main operational cost is not only the proof itself, but also the update and distribution of witnesses when the revocation set changes.

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